

Assignment : 10

Deploy a project from GitHub to EC2 by creating a new security group and user data.

Step 1: First we go to EC2 Instance then we navigate to "Security Groups" under Network & Security.

The screenshot shows the AWS Management Console for the EC2 Instances page. The left sidebar contains navigation links for Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, Images, Elastic Block Store, Network & Security, and Load Balancing. The main content area displays the 'Instances (1/1)' list with columns for Name, Instance ID, Instance state, Instance type, Status check, Alarm status, Availability Zone, and Public IP. The 'Pra-Instance' (i-0140c0933ea49d25b) is selected, and the 'Details' tab is active. The instance is in a 'Shutting-down' state. The 'Instance summary' shows the instance ID, public IPv4 address (52.205.251.129), private IPv4 address (172.31.82.186), and public IPv4 DNS (ec2-52-205-251-129.compute-1.amazonaws.com).

Step 2: Then we create security group

The screenshot shows the AWS Management Console for the Security Groups page. The left sidebar contains navigation links for EC2, Dashboard, EC2 Global View, Events, Instances, Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity Reservations, Images, Elastic Block Store, and Snapshots. The main content area displays the 'Security Groups (1/2)' list with columns for Name, Security group ID, Security group name, VPC ID, and Description. The 'launch-wizard-1' security group (sg-0f160c1cb628e6752) is selected, and the 'Details' tab is active. The security group is associated with the 'vpc-0e595d55f47fd3d5c' VPC. The 'Details' section shows the security group name, ID, description, and VPC ID.

Step 3: In Create Security group we name it as we like then give some description.

The screenshot shows the AWS Management Console for the 'Create security group' wizard. The left sidebar contains navigation links for EC2, Security Groups, and Create security group. The main content area displays the 'Create security group' form. The 'Basic details' section shows the security group name 'MyServerGroup', description 'Allows SSH access to developers', and VPC 'vpc-0e595d55f47fd3d5c'. The 'Inbound rules' section shows four rules: Custom TCP, Custom TCP, Custom TCP, and Custom TCP, all with port range 0 and source 'Anywhere-IPv4'.

Step 4: Now in Inbound rules we add rule SSH,HTTP,HTTPS,Custom TCP, where for port range we keep it default for SSH,HTTP,HTTPS and for Custom TCP we set it to 4000. We set source to "Anywhere-IPV4" 0.0.0.0/0.

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EC2 > Security Groups > Create security group

Inbound rules

Type	Protocol	Port range	Source	Description - optional	
SSH	TCP	22	Anywh... 0.0.0.0/0		Delete
HTTP	TCP	80	Anywh... 0.0.0.0/0		Delete
HTTPS	TCP	443	Anywh... 0.0.0.0/0		Delete
Custom TCP	TCP	4000	Anywh... 0.0.0.0/0		Delete

Add rule

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Outbound rules

Type	Protocol	Port range	Destination	Description - optional	
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Step 5: Then we Create Security group

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EC2 > Security Groups > Create security group

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Outbound rules

Type	Protocol	Port range	Destination	Description - optional	
All traffic	All	All	Custom 0.0.0.0/0		Delete

Add rule

Rules with destination of 0.0.0.0/0 or ::/0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

Add new tag

You can add up to 50 more tags

Cancel Create security group

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EC2 > Security Groups > sg-0b4d979032096336b - MyServerGroup

sg-0b4d979032096336b - MyServerGroup

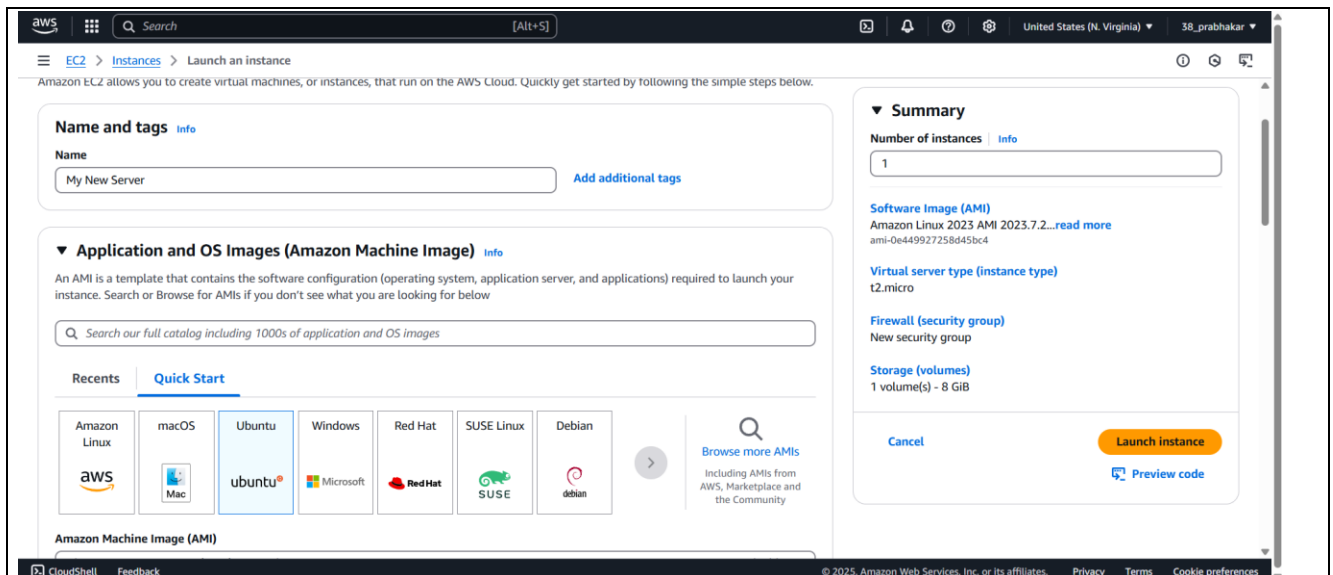
Details		VPC ID	
Security group name MyServerGroup	Security group ID sg-0b4d979032096336b	Description Allows SSH access to developers	VPC ID vpc-0e595d55f47fd3d5c
Owner 908027402686	Inbound rules count 4 Permission entries	Outbound rules count 1 Permission entry	

Inbound rules Outbound rules Sharing - new VPC associations - new Tags

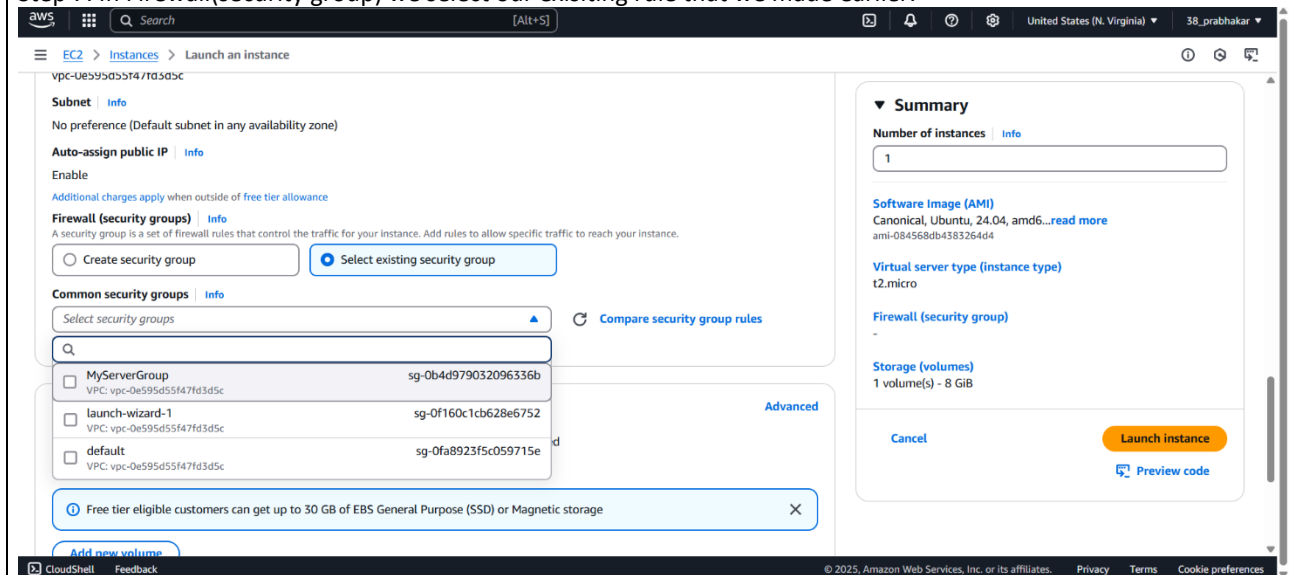
Inbound rules (4)

Name	Security group rule ID	IP version	Type	Protocol	Port range
-	sgr-0b2e6191ca62ebca2	IPv4	HTTPS	TCP	443
-	sgr-093748545978293ca	IPv4	HTTP	TCP	80
-	sgr-0c8652cb41836c6df	IPv4	Custom TCP	TCP	0
-	sgr-05111c14f3efdfeae	IPv4	SSH	TCP	22

Step 6: Now we create a EC2 instance

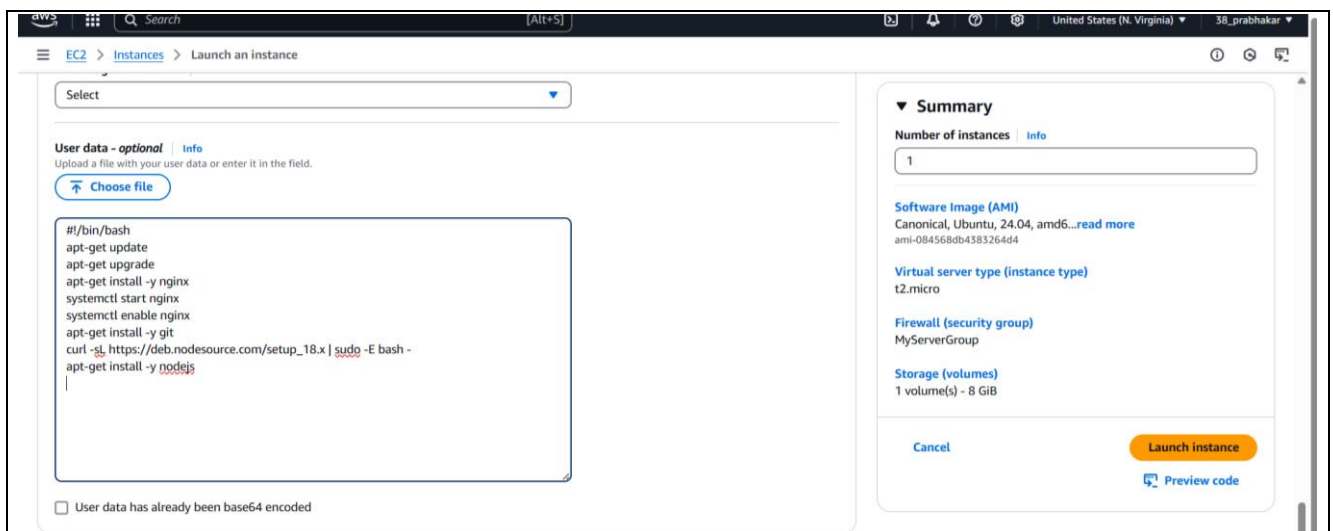


Step 7: In Firewall(security group) we select our existing rule that we made earlier.

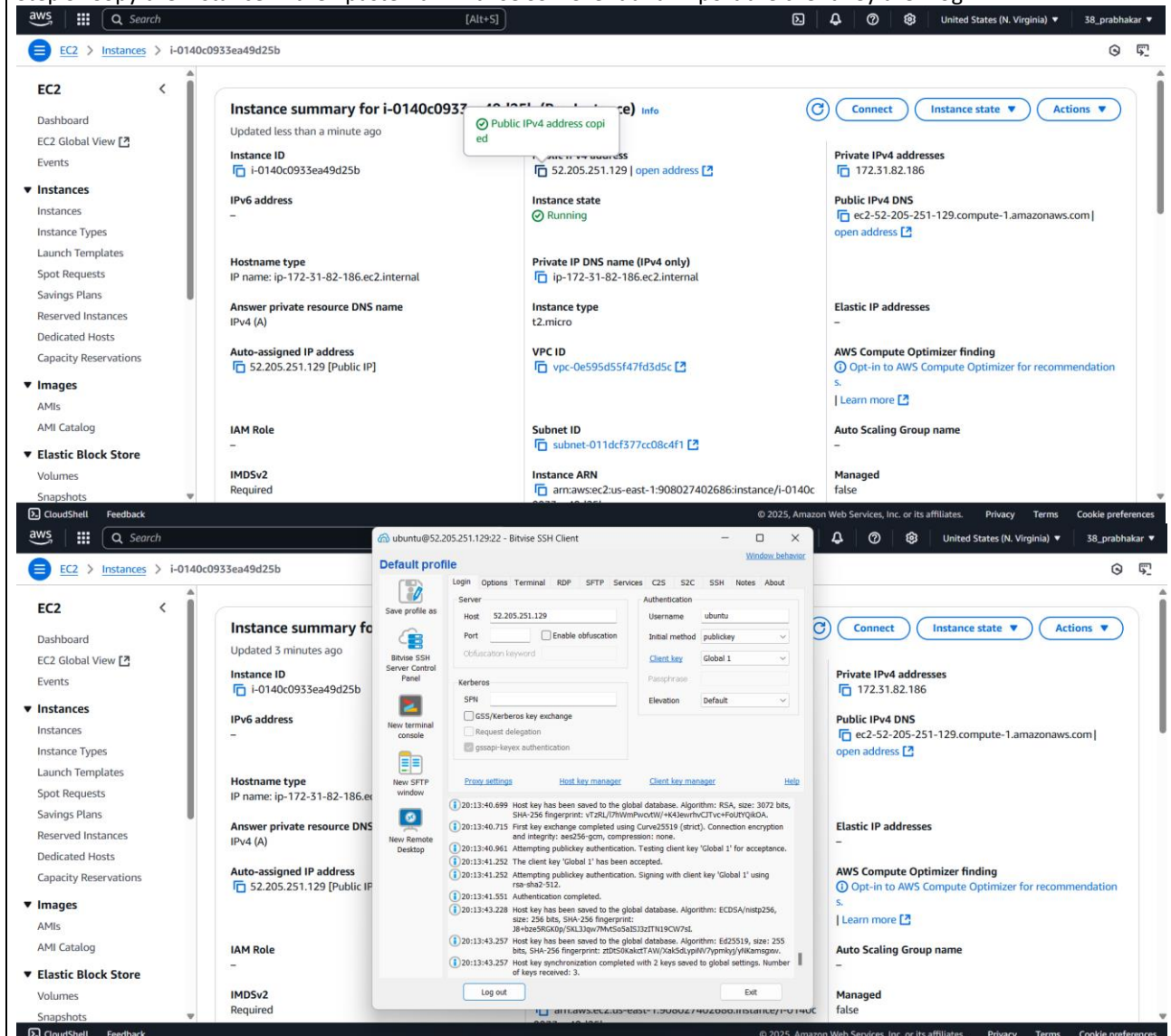


Step 8: Now we scroll down to the advance details and then in user data we write the commands

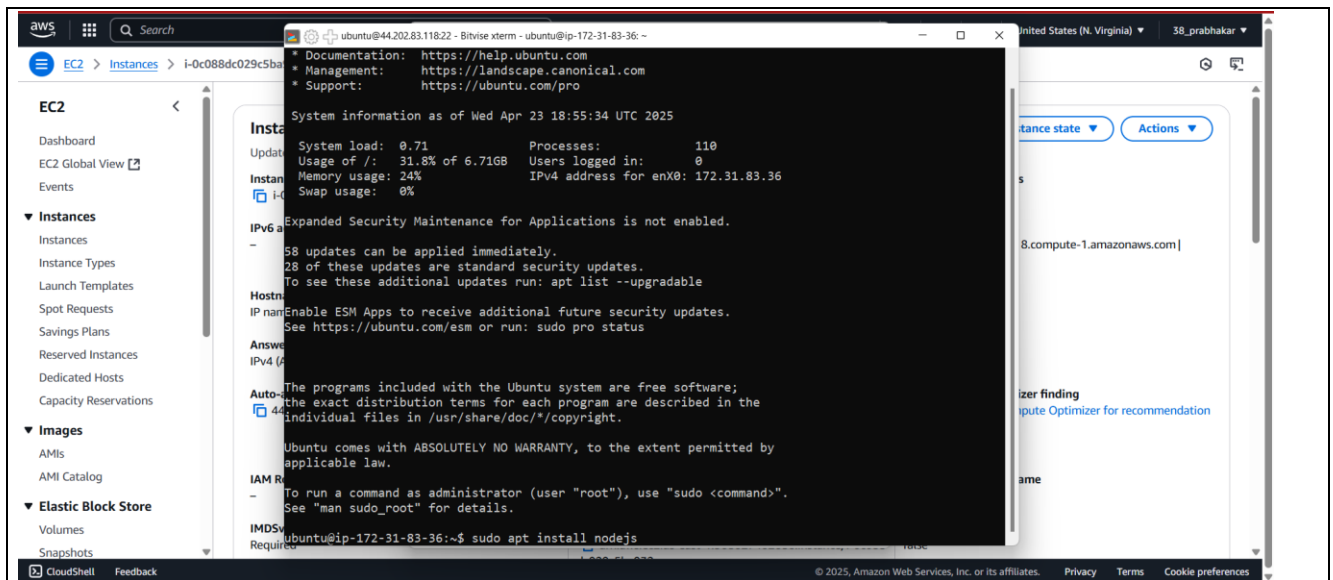
```
#!/bin/bash
apt-get update
apt-get upgrade
apt-get install -y nginx
systemctl start nginx
systemctl enable nginx
apt-get install -y git
curl -sL https://deb.nodesource.com/setup_18.x | sudo -E bash -
apt-get install -y nodejs"
And launch the instance.
```



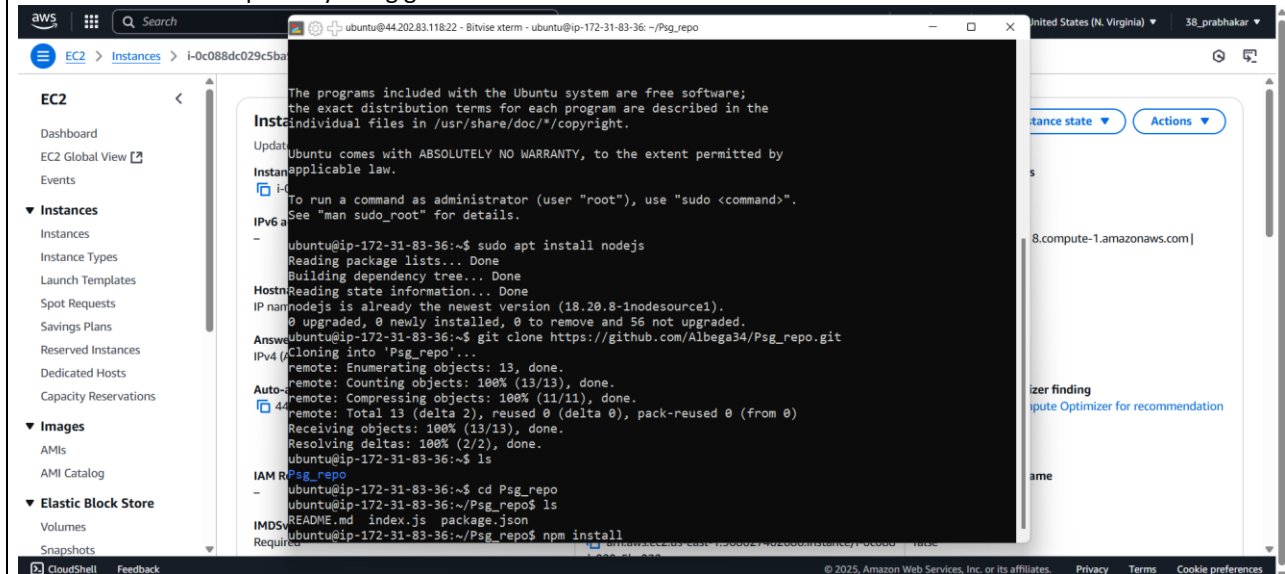
Step 9: Copy the instance IP then paster it in Bitwise SSH Client and import the client key then log in



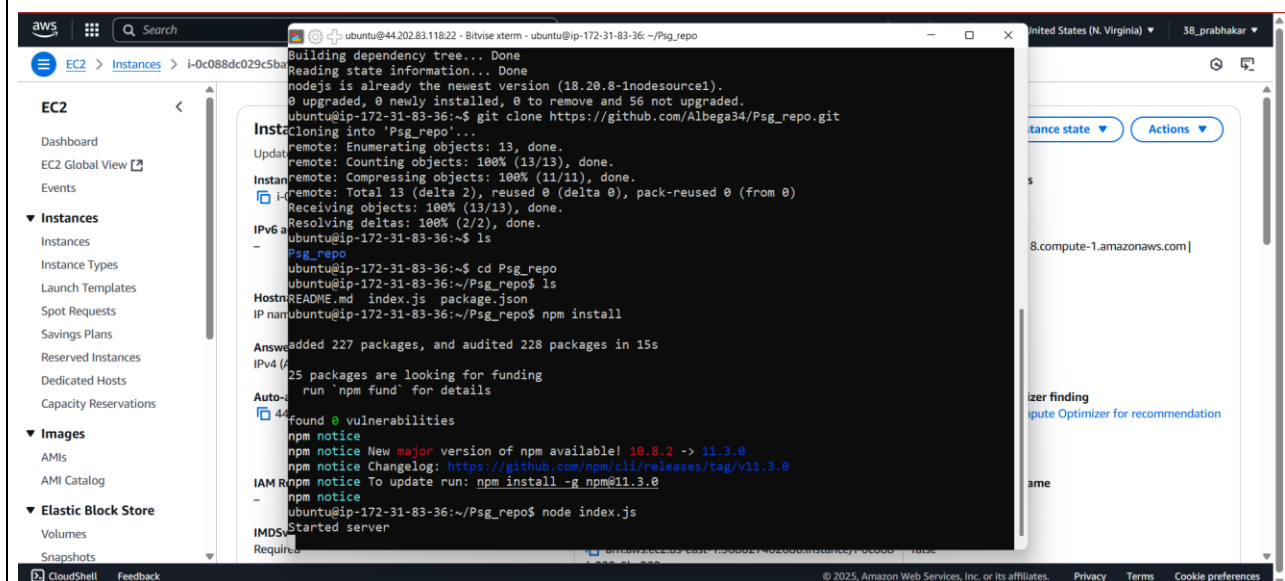
Step 10: Now that we have predefined commands we only need to run only few commands:
First command will be "sudo apt install nodejs"



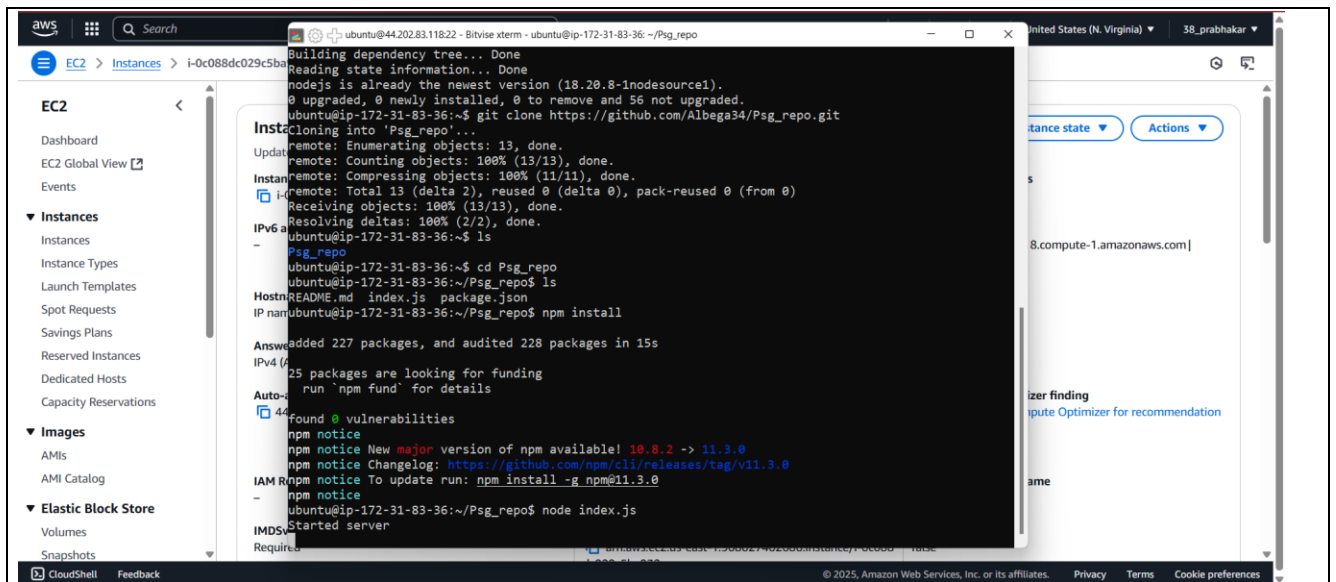
Then we clone the repository using git clone:



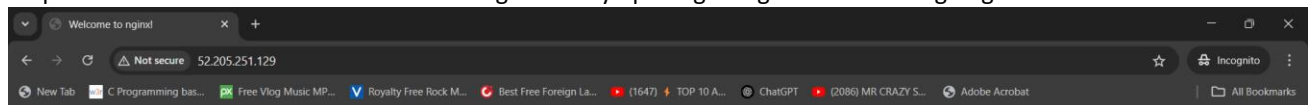
Step 11: Now we run the command "npm install"



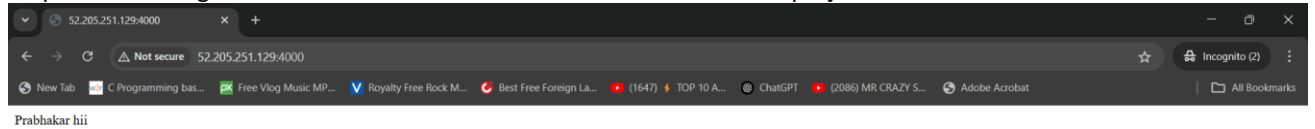
Step 13: Now we run the server using command "node index.js"



Step 14: Now we check if the server is running or not by opening incognito mode and going to the IP.



Step 15: Now we go to the IP and add :4000 at end to check our hosted project.



We have successfully hosted our project by using new security group.